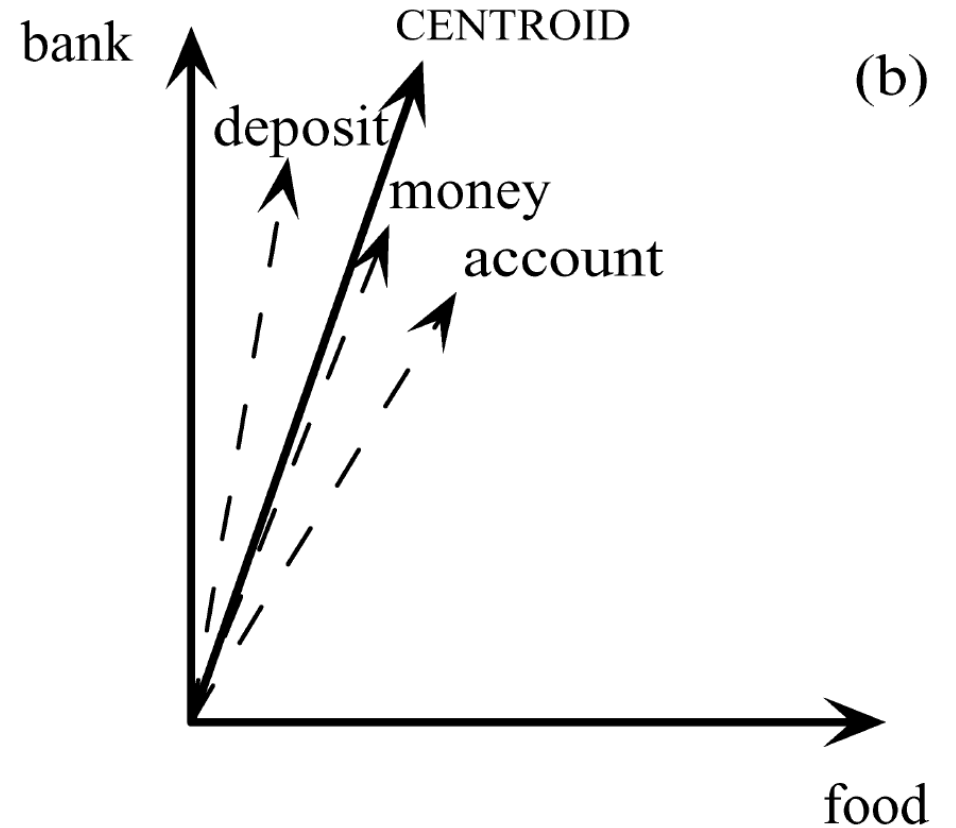
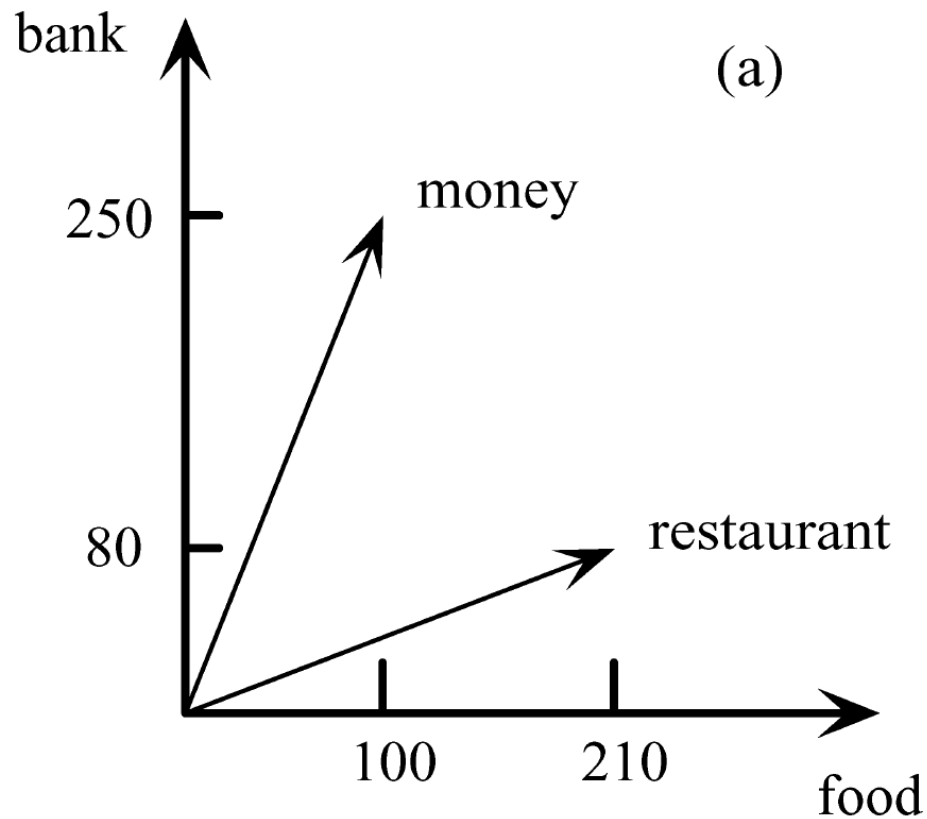


WSD : Part 2

Unsupervised WSD



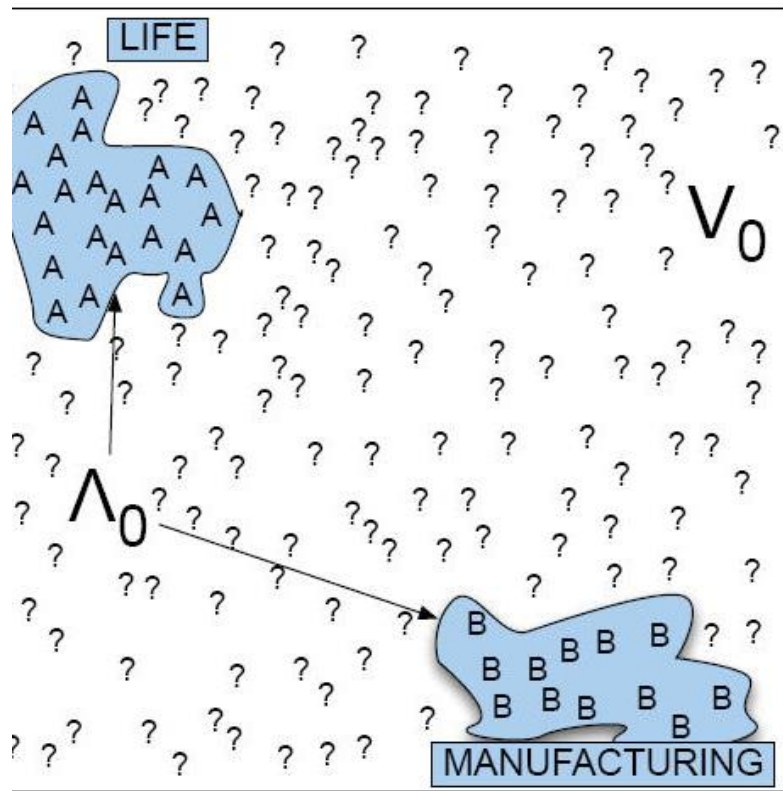
LSA, Clustering using EM or hierarchical clustering

Unsupervised WSD: Word Clustering

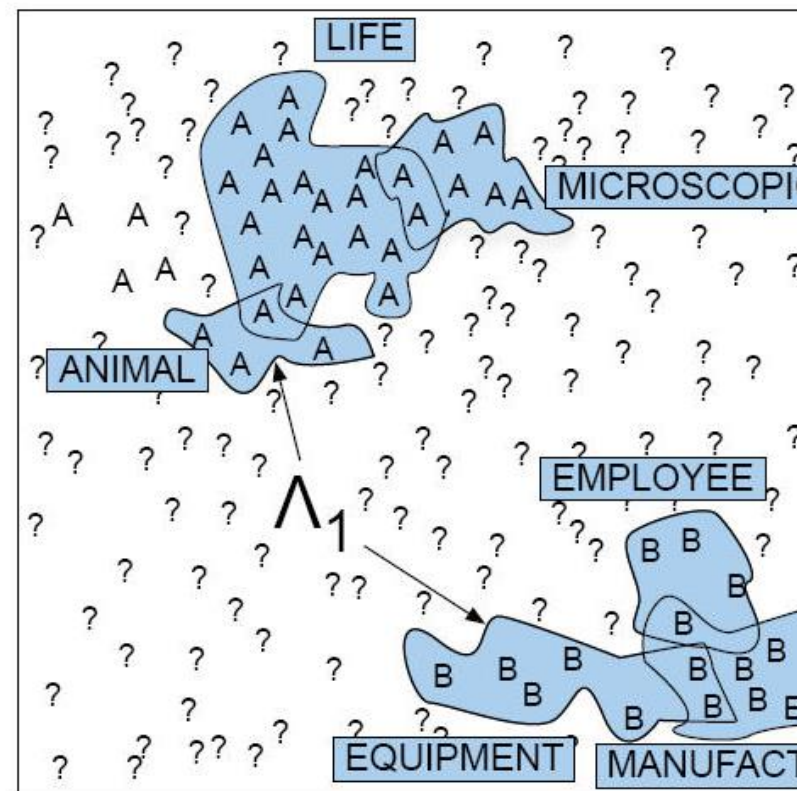
- Based on the idea of word similarity
- Distributional models of word similarity

Semi-supervised WSD

Active learning



(a)



(b)

The Web as Corpus

- An entry point into several interesting papers in NLP
- Reference paper (for those interested in NLP research):
<https://www.kilgarriff.co.uk/Publications/2003-KilgGrefenstette-WACIntro.pdf>

*Two roads diverged in a wood, and I—
I took the one less traveled by,
And that has made all the difference.*

-- (Robert Frost)

(from the book “Web as Corpus” by Maristella Gatto (Bloomsbury Academic, 2014))

Knowledge based WSD: The Lesk algorithm (using glosses)

WordNet Sense Inventory for the First Three Senses of key_n

Sense	Definition and Examples
key_n^1	Metal device shaped in such a way that when it is <i>inserted</i> into the appropriate <i>lock</i> the <i>lock</i> 's mechanism can be rotated
key_n^2	Something crucial for explaining; "the key to development is economic integration"
key_n^3	Pitch of the voice; "he spoke in a low key"

I inserted the *key* and locked the door.

Knowledge based WSD: Using selectional restrictions

Idea on selectional restrictions discussed in Part 1 of this presentation with example, Refer Section 5.2 in paper

Knowledge based WSD: Using structural methods

Self-study: Refer Section 5.3

WSD using Dominance

$$N(star_n) = \{actor_n, footballer_n, planet_n\}$$

$$sim_{dist}(star_n, actor_n) = 0.22$$

$$sim_{dist}(star_n, footballer_n) = 0.12$$

$$sim_{dist}(star_n, planet_n) = 0.08$$

$$\rightarrow \langle \underline{0.22}, \underline{0.12}, \underline{0.08} \rangle$$

$\swarrow \quad \uparrow \quad \uparrow$

Celestial object sense



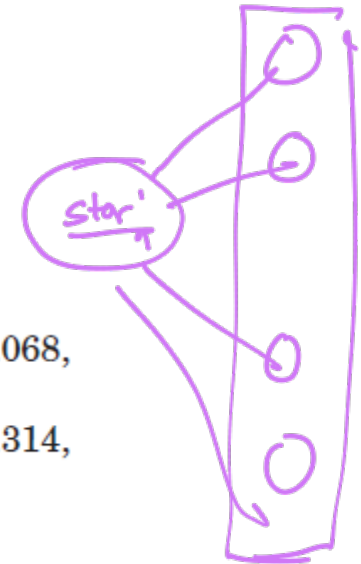
$$score_{prev}(star_n^1) = \underbrace{0.22}_{\text{actor}} \cdot \underbrace{\frac{0.01}{0.48}}_{\text{footballer}} + 0.12 \cdot \frac{0.01}{0.57} + 0.08 \cdot \frac{0.68}{0.93} = 0.068,$$

Celebrity sense



$$score_{prev}(star_n^5) = 0.22 \cdot \frac{0.42}{0.48} + 0.12 \cdot \frac{0.53}{0.57} + 0.08 \cdot \frac{0.02}{0.93} = 0.314,$$

Democracy analogy



PageRank for WSD

The church bells no longer rung on Sundays.

church

- 1: one of the groups of Christians who have their own beliefs and forms of worship
- 2: a place for public (especially Christian) worship
- 3: a service conducted in a church

bell

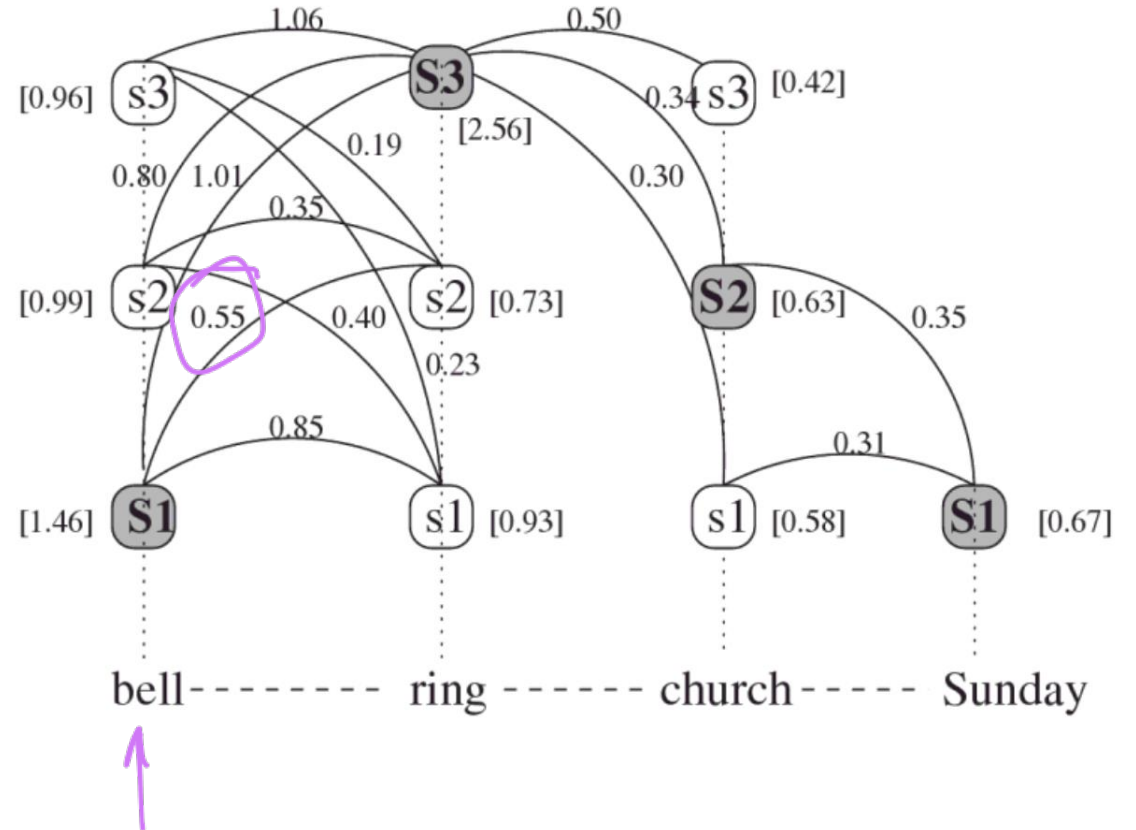
- 1: a hollow device made of metal that makes a ringing sound when struck
- 2: a push button at an outer door that gives a ringing or buzzing signal when pushed
- 3: the sound of a bell

ring

- 1: make a ringing sound
- 2: ring or echo with sound
- 3: make (bells) ring, often for the purposes of musical edification

Sunday

- 1: first day of the week; observed as a day of rest and worship by most Christians

PageRank: <https://www.aclweb.org/anthology/E09-1005.pdf>

In summarization: TextRank: <https://web.eecs.umich.edu/~mihalcea/papers/mihalcea.emnlp04.pdf>

Fig.: Recent Advances in Natural Language Processing V: Selected Papers from RANLP

edited by Nicolas Nicolov, Galia Angelova, Ruslan Mitkov

Part of Section 4.3

Lexical Chains

... the **book** that he had just taken out of the drawer. **It** was a peculiarly beautiful **book**. **Its** smooth creamy **paper** was of a kind that had not been manufactured for at least forty years past. He could guess, however, that the **book** was much older than that. He had seen **it** lying in the window of a frowsy little **junk-shop** and had been stricken immediately by an overwhelming desire to **possess** **it**. Party members were supposed not to go into ordinary **shops**. He had slipped inside and **bought** the **book** for two **dollars** fifty. Even with nothing **written** in **it**, **it** was a compromising **possession**. ¶ The thing that he was about to do was to open a **diary**. Winston fitted a **nib** into the **penholder** and sucked **it** to get the grease off. The **pen** was an archaic **instrument**, seldom used even for **signatures**, and he had **procured** one, furtively and with some difficulty, simply because of a feeling that the beautiful creamy **paper** deserved to be **written** on with a real **nib** instead of being scratched with an **ink-pencil**. Actually he was not used to **writing** by hand. He dipped the **pen** into the **ink**.

Colouring a document



Refer Section 5.3.2

WSD Evaluation

Coverage →

$$C = \frac{\text{\# answers provided}}{\text{\# total answers to provide}}$$

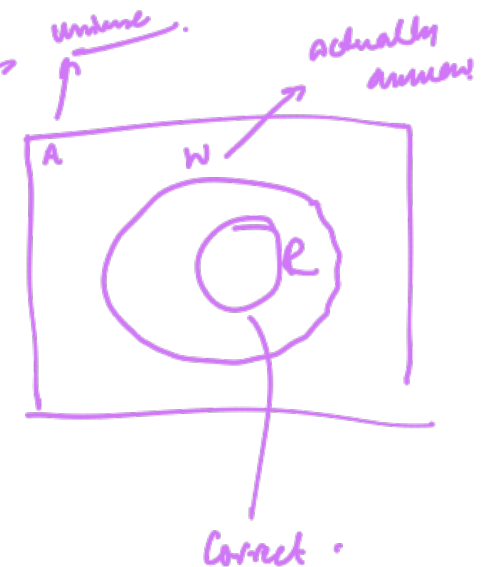
↗ $\frac{|W|}{|A|}$

↖ $\frac{|R|}{|W|}$

$$P = \frac{\text{\# correct answers provided}}{\text{\# answers provided}}$$

$$R = \frac{\text{\# correct answers provided}}{\text{\# total answers to provide}}$$

↙ $\frac{|R|}{|A|}$



$$P \geq R$$

Refer Section 7.1

Baselines, Lower and Upper Bounds

- Upper Bound : not 100% but Inter-annotator agreement

Applications of WSD and Open Problems

- Self study (gloss over and make note of important points)